

Vincent S. Pronel

732-979-9106 | vinnypronel050@gmail.com | [linkedin.com/in/vincent-pronel](https://www.linkedin.com/in/vincent-pronel) | github.com/vinnypronel

EDUCATION

Bachelor of Science in Computer Science

Kean University

Expected 2027

Union, NJ

- **Relevant Coursework:** Applied Statistics, Matrix & Linear Algebra, Discrete Structures, Data Structures, Database Management Systems, Computer Organization & Architecture, Object Oriented Analysis & Design
- **Leadership & Honors:** Dean's List, Spring 2024 and Fall 2025 | CRA UR2PhD-REU Fellow | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | NSF IUSE Research Scholar | GMiS Research Posters Competition | Web Developer, WiCyS (Women in Cyber Security)

SKILLS

AI & Evaluation: LLM API integration (Google Gemini, OpenAI), model fine-tuning, LLM-as-a-judge evaluation, prompt design, hallucination and source-fidelity measurement, human evaluation study design

Data & Analysis: SQL, MySQL, PostgreSQL, Supabase, correlation analysis, data aggregation and normalization, Google Charts, interactive dashboards

Languages: Python, Java, TypeScript, JavaScript, SQL, PHP, HTML/CSS

Engineering & Tools: Next.js, React, Node.js, REST APIs, Cloudflare (Workers, D1, R2), Vercel, Git/GitHub, Linux (Ubuntu), Claude Code, Codex

RESEARCH & ENGINEERING

AI Debate Coach Evaluation | CRA UR2PhD Research Program

Feb. 2026 – Apr. 2026

Undergraduate Researcher, Advisor: Dr. Yan Ma

Kean University

- **Built the debate application and the full evaluation pipeline and fine-tuned the model configurations under test** for a 10-week framework measuring whether prompting or fine-tuning better mitigates **LLM sycophancy** across three conditions: vanilla, prompted, and fine-tuned.
- Ran dual-track evaluation, **GPT-4-as-a-judge** scoring alongside a **12-participant human study**; measured a **100% source fabrication rate** in the fine-tuned model at triple the citation density, with **only 1 of 12 participants challenging its credibility**.
- Humans preferred the prompted model **3 to 1** while the automated judge rated the two comparably, showing **LLM-as-a-judge alone would have selected the less truthful system**; concluded that fine-tuning without retrieval grounding increases persuasiveness while degrading factuality. Presented at the **2026 UR2PhD Research Showcase**.

AI-Assisted Java Execution Visualizer | CRA UR2PhD-REU Fellowship

June 2026 – Present

Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, \$8,000 funded, 400 hours

Kean University

- **Integrated the Google Gemini API** to generate natural-language explanations of each step of **Java** program execution, designing the prompting so explanations stay grounded in the captured runtime state rather than the model's priors.
- Built and deployed the surrounding tool: a **JDI-based** tracer integration for **JDK 21**, a **JavaParser AST** layer collapsing loops and method calls into single playback steps, and a **Next.js** front end rendering variables, objects, the call stack, and the heap.
- Built the participant harness for an **IRB-approved pilot study** measuring whether **AI-assisted explanation** improves learning outcomes and reduces cognitive load against static instructional materials.

Jobs vs. Wages | NSF IUSE Scholars Academy

May 2026 – Present

NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, \$1,400 funded, 10 weeks

Kean University

- Ingested and normalized **33,000 ZIP-code records** into population-weighted state-level data marts in **MySQL** covering **52 U.S. states and territories**; implemented **correlation analysis** across four variable pairs and built an interactive dashboard to explore them.
- **Disproved the study's own premise:** unemployment explains only **14%** of variation in household income ($r = -0.38$), while college attainment ($r = 0.79$) and poverty rate ($r = -0.74$) are more than twice as predictive. Invited to the **GMiS Research Posters Competition**.

ADDITIONAL EXPERIENCE

Independent Web Developer

Jan. 2026 – Present

Freelance Full-Stack Web Development

NJ

- Founded and run a freelance practice shipping production applications for **4 business clients across 4 live sites** on **Next.js**, **TypeScript**, **PostgreSQL**, and **Cloudflare Workers**, working **agent-first with Claude Code and Codex** across the full build and deploy cycle.
- **Live sites:** deafsharkcoffee.com | daysicecream.com (**4,000+ monthly visitors**) | trainingaio.com (**delivered in 2 weeks**) | adzio.io.

Data Visualization & Analytics Intern

July 2026

Group Summer Scholars Research Program, Kean University

Union, NJ

- **Lectured daily and mentored 5 high school students** under **Dr. Ching-yu Huang**, authoring the curriculum that took every student from raw datasets to a published **database-driven website** and a research poster presented under my guidance.

Computer Science Tutor (Code Samurai)

Jan. 2026 – Present

Kean University

Union, NJ

- **Selected by CS faculty** to tutor **Database Management Systems** in twice-weekly one-on-one sessions for **50+** undergraduates; coach students through **relational database design** and writing efficient **SQL** queries.